

to monitor and control incoming and outgoing network traffic. pfSense, OPNsense, and WireGuard are three powerful open source tools that provide comprehensive network security features.

- **pfSense:** pfSense is a free, open source firewall and router platform based on FreeBSD. It offers advanced security features like VPN support, load balancing, and intrusion detection.
- **OPNsense:** Similar to pfSense, OPNsense is another open source firewall and routing platform based on FreeBSD. It includes features such as intrusion prevention (IPS), traffic shaping, and support for multiple VPN protocols.
- **WireGuard:** WireGuard is a lightweight and high-performance VPN protocol that focuses on simplicity and speed. It is often integrated into firewalls and routers, providing secure remote access for employees or connecting remote sites securely.

Detecting and preventing

unauthorised access with IDS and IPS:

Intrusion detection systems (IDS) and intrusion prevention systems (IPS) are designed to detect and block malicious activities and unauthorised access attempts within a network. Tools like Snort, Suricata, and Zeek (formerly known as Bro) provide open source solutions for real-time traffic analysis and threat detection.

- **Snort:** Snort is one of the most popular open source IDS/IPS tools. It can be used to detect a wide range of attacks, including denial of service (DoS), buffer overflows, and SQL injections.
- **Suricata:** Suricata is an advanced IDS/IPS tool that can handle high-speed networks and offer real-time traffic analysis. It is multi-threaded, allowing it to scale and analyse large volumes of network data. Suricata's multi-protocol support makes it a valuable tool for detecting complex

network attacks.

- **Zeek (formerly Bro):** Zeek is an open source network monitoring tool that focuses on network traffic analysis and security monitoring. Unlike traditional IDS/IPS tools, Zeek focuses on the network context and provides detailed logging of network sessions.

Monitoring network traffic in

real time: Network traffic analysis plays a crucial role in identifying unusual or malicious behavior. Open source tools like Wireshark, ntopng, and Nagios provide comprehensive monitoring solutions that allow network administrators to capture and analyse traffic patterns and detect potential threats.

- **Wireshark:** Wireshark is a widely used open source network protocol analyser. It allows administrators to capture and inspect the packets flowing through their network in real time.
- **ntopng:** ntopng is a high-performance network monitoring tool that provides real-time traffic analysis and visualisation. It supports flow-based monitoring, which gives network engineers a better understanding of traffic patterns, bandwidth usage, and potential bottlenecks.
- **Nagios:** While Nagios is primarily known for system monitoring, it also provides capabilities for network traffic monitoring. By integrating with other tools like Wireshark or ntopng, Nagios can become a comprehensive network monitoring solution.

Identifying network weaknesses

with vulnerability scanning:

Vulnerability scanning is another critical component of network security, helping to identify weaknesses in your network infrastructure before attackers can exploit them. Open source tools like OpenVAS, Nikto, and Nessus are invaluable for conducting proactive vulnerability assessments.

- **OpenVAS:** OpenVAS is a powerful open source vulnerability scanner that performs thorough scans on network devices, servers, and applications.
- **Nikto:** Nikto is a web server scanner that focuses on detecting vulnerabilities within web applications. It scans for common web server issues, such as insecure headers, outdated software, and missing security patches.
- **Nessus:** Nessus is a widely used vulnerability scanner. While its commercial version is well-known, there is also an open source version available for individual use.

DevSecOps: Integrating security into the DevOps pipeline

DevSecOps is the practice of integrating security into the DevOps pipeline, ensuring that security measures are automated and enforced throughout the software development and deployment process. Traditional DevOps focuses on speeding up development and operations, but it may neglect security, leaving vulnerabilities exposed. DevSecOps bridges this gap by incorporating security practices into the CI/CD (continuous integration/continuous deployment) pipeline, ensuring that security is built in. Key principles of DevSecOps include:

Shift left: Security should be integrated as early as possible in the development lifecycle. This approach, known as 'shift left', encourages developers to identify and address vulnerabilities during the design and coding stages, rather than waiting until the testing or production phase.

Automation of security testing:

By automating security checks, organisations can ensure that security is continuously tested and validated throughout the development pipeline. This eliminates the need for manual checks and reduces the risk of